GridSkip
From Studio Right-to-Repair Experiment to an Interactive 3D Web Showcase & Build Guide

Documentation

gridskip.vercel.app

Rajatkumar Shahare
MA Experience Design
251530740017

From studio to seminar

Modern electronics are often closed systems that depend on company-controlled software, cloud services, and mobile networks. This means companies can control how people use their devices and communicate. GridSkip was created in the design studio as an alternative: a small, battery-powered text communicator that works without the internet, cellular networks, or cloud services. It uses 433 MHz LoRa radio to send messages directly, allowing people to communicate independently.

I realized that open-source hardware has little impact if people can only see it in an exhibition. So, the project moved into its second phase: creating a user-friendly, hands-on website and build guide that helps anyone, anywhere, source parts, 3D print the case, connect the wires, install the software, and build their own mesh communicator.



Rather than relying on flat photography, the web build guide opens with a dynamic Three.js WebGL CAD explosion. Visitor can orbit 360° and hover over discrete components to inspect.

01 / WHY

If you can't repair it, you don't fully own it.

Most devices have more than one owner: the company, the software, the repair centre, sometimes the government. GridSkip began as a study of the right to repair and became a question about control. If someone else decides how our devices work, they also decide how we communicate.

GridSkip is a small answer. Every part is visible, every wire is traceable, and the firmware is yours to change.

Read Reclaiming Repair →

02 / HOW IT WORKS

A private mesh of your own.

An ESP32 drives a LoRa radio, a small OLED, a rotary encoder, and two buttons, all inside a 3D-printed handheld case. The firmware is a modified Meshtastic build that only talks to GridSkip nodes sharing your channel key. Messages are encrypted end-to-end.

HARDWARE MODULE	FUNCTION
ESP32 DevKit V1	the brain
RA-02 (SX1278)	the 433 MHz radio
1.3″ OLED	the screen
Rotary encoder + 2 buttons	the input
LiPo + TP4056 + boost	power and charging

Have added an web gl emulator that runs same ui and controls of the device

03 / USE IT

One dial, two buttons.

TURN Scroll through options or pick a character	PRESS Select the highlighted item or confirm
BACK / DELETE Go back one screen or delete a character	SEND Transmit the composed message to your channel



STEPS

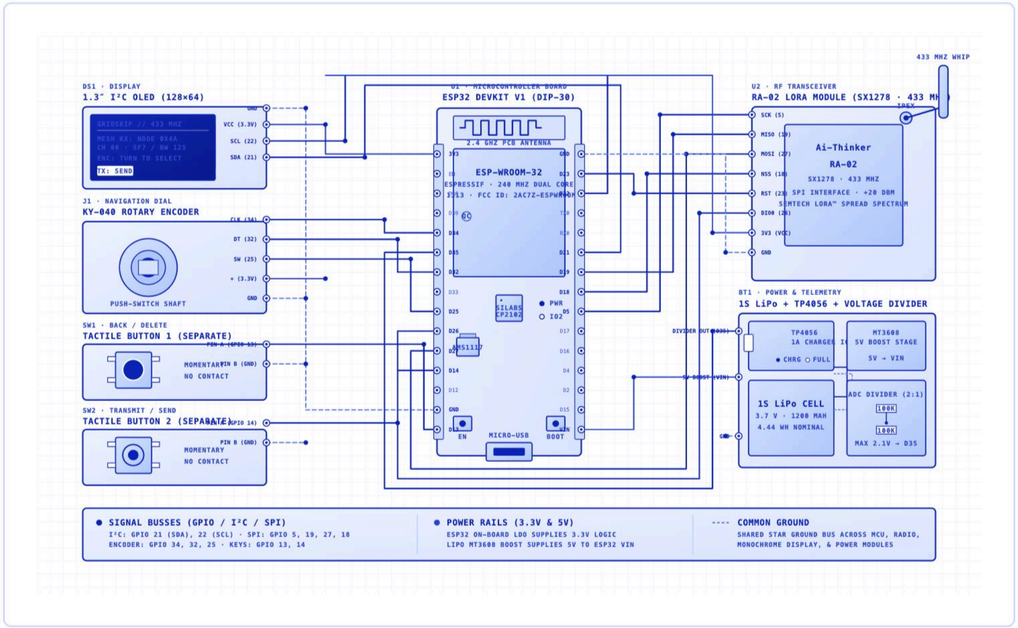
1. Order the parts and 3D-print the enclosure case.
   - ▼ Front.stl (238 KB)
   - ▼ Back.stl (81 KB)
   - 📦 Download All STLs (.ZIP)
2. Wire everything using the schematic below.
3. Flash the firmware. [Gridskip-Firmware repository and flash steps ↗]
4. Set the same key on both nodes.
5. Close the case, power both on, and send "hello."

> WARNING
> **The RA-02 is 3.3 V only. 5 V on its VCC will destroy it.**

Have added links for 3d enclosure and firmware

WIRING SCHEMATIC

The RA-02 runs at 3.3 V — never connect its VCC to 5 V.



Wiring schematic is still broken i tried to make it with AI but its not able to build it properly i will draw and update the code once i get some free time.